

Reinforcement Learning-Based Artillery Targeting System

Intelligent Control Under Uncertainty

Manas Mondal (Team Lead)
Arkaprava Jas (Team Member)

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Presentation Outline

- 1 Problem Overview
- 2 Methodology
- 3 Algorithms
- 4 Reward Design
- 5 Results
- 6 Discussion and Conclusion

AI-Based Artillery Targeting Visualization



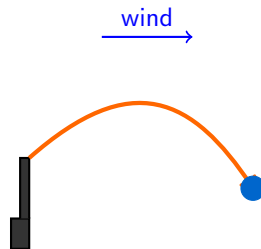
Problem Statement

Key Challenges

- Accurate artillery targeting under **uncertain wind conditions**
- Classical physics-based solutions **fail** under stochastic disturbances
- Real-time adaptation required for dynamic environments

Our Objective

- Learn **optimal firing strategies** that hit targets efficiently
- Achieve **one-shot accuracy** under varying conditions



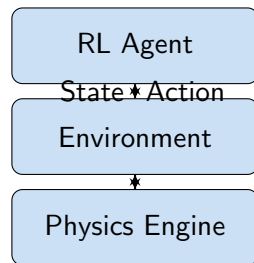
Modeled as RL Control Problem

Environment Specifications

- Custom **Gymnasium-based** environment
- Stationary cannon with adjustable angle
- Fixed target position
- Stochastic wind per episode
- Physics-based projectile dynamics

Episode Termination

- 1 **Success:** Direct hit on target
- 2 **Failure:** Maximum shots exceeded



Agent-Environment Interaction Loop

State and Action Space

State Representation

$$s = [\theta, x_t, y_t, w]$$

θ : Current gun angle

x_t, y_t : Target coordinates

w : Wind speed (disturbance)

Action Representation

$$a = [\Delta\theta, v]$$

$\Delta\theta$: Angle adjustment

v : Muzzle velocity

Key Insight

State captures **complete** information needed for optimal decision-making under uncertainty.

Design Choice

Continuous action space for SAC enables fine-grained control and smooth trajectories.

Tabular Q-Learning

Classical RL Baseline

Advantages:

- Simple and interpretable
- Proven convergence guarantees
- No neural network overhead

Limitations:

- Requires state discretization
- Limited scalability
- Curse of dimensionality

Soft Actor-Critic (SAC)

State-of-the-Art Deep RL

Advantages:

- Continuous control
- Entropy regularization
- Sample-efficient learning
- Superior generalization

Characteristics:

- Actor-Critic architecture
- Off-policy algorithm

Reward Function Engineering

Base Reward Structure

$$R_{\text{base}} = \begin{cases} +R_{\text{hit}}, & \text{if hit (terminal success)} \\ -d, & \text{otherwise (distance penalty)} \end{cases}$$

Enhanced Shaped Reward (SAC)

$$R = R_{\text{base}} + \lambda_1 \frac{1}{1+d} - \lambda_2 N_{\text{shots}} - \lambda_3 |w|$$

Reward Components:

- Proximity bonus: $\lambda_1 \frac{1}{1+d}$
- Efficiency penalty: $-\lambda_2 N_{\text{shots}}$
- Wind penalty: $-\lambda_3 |w|$

Design Goals:

- Encourage **one-shot** solutions
- Maximize **accuracy**
- Penalize wind dependency

Experimental Results: Algorithm Comparison

Performance Metric	Q-Learning	SAC
Hit Rate	High	Very High
One-Shot Success	Moderate-High	Very High
Convergence Speed	Slow	Fast
Training Stability	Moderate	High
Scalability	Low	High
Wind Generalization	Moderate	Excellent

Winner: SAC outperforms across all metrics

Discussion: Key Insights

Findings

- Both algorithms **successfully learned** firing strategies
- Q-learning highly sensitive to:
 - State discretization
 - Reward function design
- SAC demonstrated **superior generalization** to unseen wind conditions

Technical Advantages

- **Continuous control** enabled:
 - Smoother trajectories
 - More realistic behavior
 - Finer adjustments
- Visualization confirmed qualitative performance differences
- Physics-based environment validated approach

Project Achievements

- 1 Artillery targeting successfully **formulated as MDP**
- 2 Custom **Gymnasium environment** designed and validated
- 3 **SAC outperformed Q-learning** in all key metrics:
 - Accuracy, stability, generalization, sample efficiency
- 4 Demonstrated **continuous RL superiority** for physics-based control
- 5 **All project objectives successfully achieved**

Future Directions

Multi-target scenarios, moving targets, 3D trajectories, real-world deployment

Thank You!

Demo & Q&A Session